APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Fourth Semester B.Tech Degree Examination July 2021 (2019 Scheme)

Course Code: EET204
Course Name: ELECTROMAGNETIC THEORY

Max. Marks: 100 Duration: 3 Hours

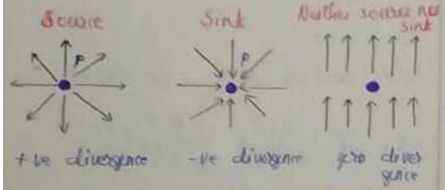
PART A

(Answer all questions; each question carries 3 marks)

SL.NO	ANSWERS
1	

Divergence of a Vector Field

The divergence of $\vec{A}$ at a given point 'P' is the outward flux per unit volume as the volume shrinks about P.



$$\text{div}\,\vec{A} = \nabla\cdot\vec{A} = \lim_{\Delta V\to 0} \frac{\oint_S \vec{A}\cdot\vec{ds}}{\Delta V}$$

where ΔV is the volume enclosed by the closed surface 'S' in which 'P' is located.

Physically, we may regard the divergence of vector field $\vec{A}$ at a given point as a measure of how much the field diverges from that point.

If the source field spreads out the net flow of flux is outwards. Hence divergence is +ve.

If the system has inward flow of flux (sink) then the divergence will be -ve

If the system behaves neither source nor sink, it has zero divergence.

For cartesian, $\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z$

$$\nabla\cdot\vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

For cylindrical, $\vec{A} = A_\rho\hat{a}_\rho + A_\phi\hat{a}_\phi + A_z\hat{a}_z$

$$\nabla\cdot\vec{A} = \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho A_\rho) + \frac{1}{\rho}\frac{\partial A_\phi}{\partial\phi} + \frac{\partial A_z}{\partial z}$$

For spherical $\vec{A} = A_r\hat{a}_r + A_\theta\hat{a}_\theta + A_\phi\hat{a}_\phi$

$$\nabla\cdot\vec{A} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2A_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta A_\theta) + \frac{1}{r\sin\theta}\frac{\partial A_\phi}{\partial\phi}$$

Properties

It produces a scalar field

The divergence of a scalar r, does makes no sense

$\nabla\cdot(\vec{A}+\vec{B}) = \nabla\cdot\vec{A} + \nabla\cdot\vec{B}$

$\nabla\cdot(V\vec{A}) = V\nabla\cdot\vec{A} + \vec{A}\cdot\nabla V$

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$$V = \frac{3z}{\rho}\cos\phi$$

$$\nabla V = \frac{\partial V}{\partial \rho}\hat{a}_\rho + \frac{1}{\rho}\frac{\partial V}{\partial \phi}\hat{a}_\phi + \frac{\partial V}{\partial z}\hat{a}_z$$

$$= \frac{-3z}{\rho^2}\cos\phi\,\hat{a}_\rho + \frac{-3}{\rho}z\sin\phi\,\hat{a}_\phi + \frac{3}{\rho}\cos\phi\,\hat{a}_z$$

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The Equipotential Surfaces

The surface which is the locus of all points which are at the same potential is known as the equipotential surface.

No work is required to move a charge from one point to another on the equipotential surface.

ie; if a point charge is moved from point A to point B, on an equipotential surface, then the work done in moving the charge is given by

$$W = q(V_A - V_B)$$

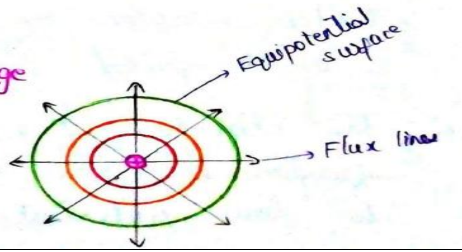
where V_A and V_B are the potentials at points A & B respectively.

Since the surface is equipotent then $V_A = V_B$

So $V_A - V_B = 0$

$W = 0$

Point charge



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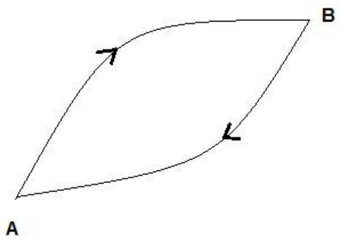
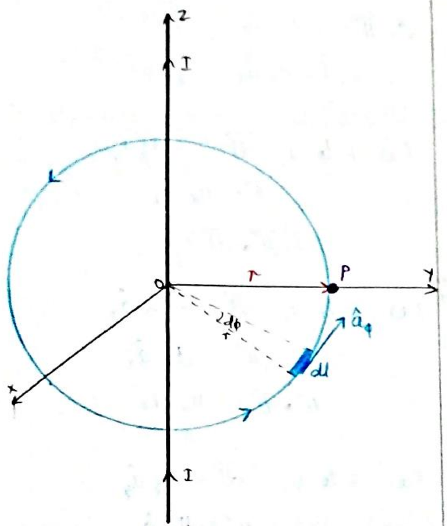
4	A force is said to be conservative if the work done by the force in moving a particle from one point to another point depends only on the initial and final points and not on the path followed. The field where the conservative force is felt is called conservative field. The electric field is a conservative field. i.e.Consider an electric field due to a charge Q. The work done to carry a test charge q from one point A to another point B in the field due to Q does not depend upon the path followed. It depends only upon the initial and final positions A and B. It will be independent of the path followed. So we say that the electric field is conservative in nature. Conservative nature of electric field means that the line integral of the electric field along a closed path is zero. Consider two point A and B in an electric field due to a charge Q. Work is done by the electric field in moving a test charge from point A to B along a closed path. For that, we have to add up the work done for all the infinitesimal segments into which the path A and B is divided. Such an integral evaluated along a line is called a line integral.  *Let us consider the line integral of electric field along A to B* : $\int_A^B \vec{E}.\vec{dl}$ *The line integral along path B to A* : $\int_B^A \vec{E}.\vec{dl}$ *The line integral along the closed path will be*: $\int_A^B \vec{E}.\vec{dl} + \int_B^A \vec{E}.\vec{dl}$ *But* $\int_B^A \vec{E}.\vec{dl} = -\int_A^B \vec{E}.\vec{dl}$ $\Rightarrow \int_A^B \vec{E}.\vec{dl} + \int_B^A \vec{E}.\vec{dl} = \int_A^B \vec{E}.\vec{dl} + -\int_A^B \vec{E}.\vec{dl} = 0$ i.e.Line integral of an electric field along a closed path is zero means that the work done by the electric field is independent of the path and depends only on the endpoints A and B of integration.

5		

Applications

i Magnetic field intensity due to an infinite conductor



Consider an infinitely long straight conductor carrying direct current I placed along z axes. Consider a closed circular path of radius r, known as Amperian loop which encloses the straight conductor. Point 'P' is at a distance r from the conductor.

According to Ampere's law

$$\oint_L \vec{H} \cdot \vec{dl} = I$$

$\vec{dl} = r\,d\phi\,\hat{a}_\phi$

$\vec{H} = H_\phi\,\hat{a}_\phi$ → tangential component

$\vec{H} \cdot \vec{dl} = r H_\phi\, d\phi$

$$\oint_L \vec{H} \cdot \vec{dl} = \int_0^{2\pi} r H_\phi\, d\phi$$

$$= r H_\phi \int_0^{2\pi} d\phi$$

$$= r H_\phi\, 2\pi$$

ie; $I = 2\pi r H_\phi$

$$H_\phi = \frac{I}{2\pi r}$$

$$\vec{H} = \frac{I}{2\pi r}\,\hat{a}_\phi$$

ii Magnetic Field Intens[ity] due to infinite Sheet Current

Consider an infinite current sh[eet] in the $z=0$ plane. If the sheet [has] a uniform current density $\vec{K} = k_y\,\hat{a}_y$ A/m [current is fl[owing] in +ve y direction].

We regard the infinite sheet [as] comprising of filaments.

6	

Laplace equation for Scalar Magnetic Potential V_m

As monopole of magnetic field does not exist, therefore;

$$\nabla \cdot \vec{B} = 0$$

$$\Rightarrow \nabla \cdot (\mu \vec{H}) = 0$$

$$\Rightarrow \nabla \cdot \vec{H} = 0$$

$$\Rightarrow \nabla \cdot (-\nabla V_m) = 0$$

$$\Rightarrow \nabla^2 V_m = 0, \; (\vec{J} = 0)$$

Vector Magnetic Potential $\vec{A}$

For a magnetostatic field,

$$\nabla \cdot \vec{B} = 0$$

but $\vec{B} = \nabla \times \vec{A}$

$$\cancel{\Rightarrow \nabla \cdot (\nabla \times \vec{A}) = 0}$$

ie; The curl of the vector magnetic potential gives magnetic flux density.

From Ampere Circuital law,

$$\nabla \times \vec{H} = \vec{J}$$

$$\nabla \times \left(\frac{\vec{B}}{\mu}\right) = \vec{J}$$

$$\nabla \times \vec{B} = \mu \vec{J}$$

$$\nabla \times (\nabla \times \vec{A}) = \mu \vec{J}$$

Using properties of curl,

$$\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$$

ie; $\nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu \vec{J}$

$$\vec{J} = \frac{1}{\mu}\left[\nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}\right]$$

$$\vec{J} = \frac{1}{\mu}\left[\nabla \times (\nabla \times \vec{A})\right]$$

Poissons equation for Vector Magnetic Potential $\vec{A}$

Assume that $\nabla \cdot \vec{A} = 0$

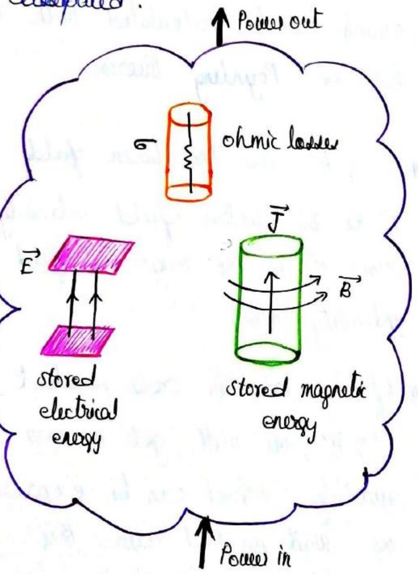
then, $J = \frac{1}{\mu}(-\nabla^2 \vec{A})$

$$\nabla^2 \vec{A} = -\mu \vec{J}$$

7		Using EM waves, energy can be transported from transmitter to receiver. The energy stored in electromagnetic fields is transmitted at a certain rate of energy flow which can be calculated with the help of Poynting theorem. # $\vec{E}$ & $\vec{H}$ are the basic fields. $\vec{E}$ is the electric field intensity (V/m), $\vec{H}$ is the magnetic field intensity (A/m). # If we take the cross product of $\vec{E}$ & $\vec{H}$, we will get a new quantity which can be expressed as Watt per Unit area. This quantity is called Power Density Power density, $\vec{P} = \vec{E} \times \vec{H}$ ↓ Poynting vector

The Poynting Theorem states that is based on Law of Conservation of energy in Electromagnetism.

Poynting theorem states that
"The net power flowing out of a given volume V is equal to the time rate of decrease in the energy stored within the volume V minus the ohmic power dissipated".



Suppose $\vec{E} = E_x \hat{a}_x$, $\vec{H} = H_y \hat{a}_y$

$$\vec{P} = \vec{E} \times \vec{H}$$
$$= E_x \hat{a}_x \times H_y \hat{a}_y$$
$$= P_z \hat{a}_z$$

$\vec{E}$, $\vec{H}$ & $\vec{P}$ are mutually perpendicular

$\vec{P}$ is called Poynting vector. $\vec{P}$ is the instantaneous power density vector associated with EM field at a given point.

To get a net power flowing out any surface, $\vec{P}$ is integrated over the same closed surface.

Integral Form of Poynting Theorem

$$\underbrace{\oint_S (\vec{E} \times \vec{H}) \cdot \vec{ds}}_{\text{Total power leaving the volume}} = \underbrace{-\frac{\partial}{\partial t} \int_V \left[\frac{1}{2} \varepsilon E^2 + \frac{1}{2} \mu H^2\right] dV}_{\text{Rate of decrease in energy stored in EM field}} \underbrace{- \int_V \sigma E^2 dV}_{\text{ohmic power dissipated}}$$

Proof

Let's consider Maxwell's eqn based on Ampere's Circuital law.

$$\nabla \times \vec{H} = \vec{J_c} + \frac{\partial \vec{D}}{\partial t}$$

$$\vec{J_c} = (\nabla \times \vec{H}) - \frac{\partial \vec{D}}{\partial t} \quad \text{---} (A)$$

8

For good conductor,

$$\vec{E} = E_0 e^{-\alpha z} \cos(\omega t - \beta z)\, \hat{a}_x$$

$$\vec{H} = \frac{E_0}{\eta} e^{-\alpha z} \cos(\omega t - \beta z - \theta_n)\, \hat{a}_y$$

$$\vec{H} = \frac{E_0}{\sqrt{\frac{\mu\omega}{\sigma}}} e^{-\alpha z} \cos(\omega t - \beta z - 45^\circ)\, \hat{a}_y$$

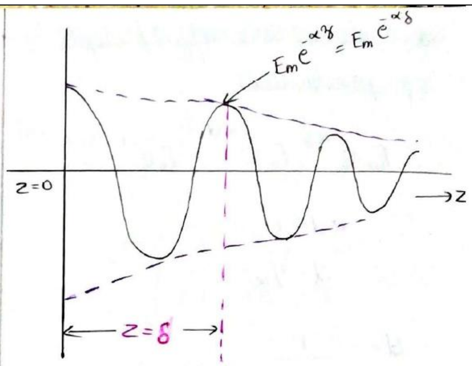
Skin depth (Skin effect) (Depth of penetration)

Consider the component of electric field E_x travelling in +ve z direction. When it travels in good conductor, the conductivity is high & the attenuation constant α is also high.

We can write such a component in phasor form as

$$E_x = E_m e^{-\alpha z} e^{-j\beta z} \quad \ldots (1)$$

Such a wave is shown below



At z=0, amplitude of the component E_x is E_m, while at $z = \delta$, amplitude is $E_m e^{-\alpha\delta}$.

In distance $z = \delta$, the ~~amplitud~~ amplitude gets reduced by a factor $e^{-\alpha\delta}$.

If we select $\delta = 1/\alpha$, then the factor becomes $e^{-1} = 0.368$.

ie; Over a distance $\delta = 1/\alpha$, the amplitude of the wave decreases approximately 37% of original value.

The distance through which the amplitude of the ~~wave~~ travelling wave decreases to 37% of the original amplitude is called

skin depth/ skin effect/ depth of penetration:

$$E_0 e^{-\alpha z} \quad E_0 e^{-\alpha\delta} = E_0 e^{-1}$$

$$\alpha\delta = 1$$

$$\delta = 1/\alpha$$

$$\delta = \frac{1}{\sqrt{\frac{\mu\sigma\omega}{2}}}$$

$$\boxed{\delta = \sqrt{\frac{2}{\mu\sigma\omega}}}$$

Surface Resistance (R_s)

Surface resistance is the real part of intrinsic impedence η for a good conductor.

$$\eta = \sqrt{\frac{\omega\mu}{\sigma}} \angle 45^\circ \quad \ldots (1)$$

$$\alpha = \sqrt{\frac{\omega\mu\sigma}{2}} \quad \ldots (2)$$

$$\delta = \frac{1}{\alpha} \quad \ldots (3)$$

$$\alpha = \sqrt{\frac{\omega\mu}{\sigma}\,\frac{\sigma^2}{2}} =$$

$$\alpha = \eta\sqrt{\frac{\sigma^2}{2}}$$

$$\alpha = \eta\frac{\sigma}{\sqrt{2}}$$

$$\eta = \frac{\alpha\sqrt{2}}{\sigma} \angle 45^\circ$$

$$\eta = \frac{\alpha\sqrt{2}}{\sigma} e^{j45^\circ}$$

$$\eta = \frac{\alpha\sqrt{2}}{\sigma}(\cos 45^\circ + j\sin 45^\circ)$$

$$\eta = \frac{\alpha\cancel{\sqrt{2}}}{\sigma}\frac{1}{\cancel{\sqrt{2}}} + j\frac{\alpha\cancel{\sqrt{2}}}{\sigma}\frac{1}{\cancel{\sqrt{2}}}$$

$$\eta = \frac{\alpha}{\sigma} + j\frac{\alpha}{\sigma}$$

$$\eta = \frac{1}{\sigma\delta} + j\frac{1}{\sigma\delta}$$

$$\cancel{\eta = \frac{1+j}{\sigma\delta}}$$

$$\eta = R_s + jX_s$$

Surface resistance,

$$\boxed{R_s = \frac{1}{\sigma\delta} = \text{DC resistance of conductor}}$$

$R_{ac} >> R_{dc}$ at high freq

9

EMI stands for electromagnetic interference. EMI is a common phenomenon in our world, and it is getting worse as a result of the expanse and advances of modern electronics. Since World War 2, EMI has been a military problem, but since then the problem has spread to everyday consumers and businesses. EMI shields and EMI gaskets are used to reduce the amount of EMI that leaks from an electronic device. Regardless of where EMI comes from, it will always have a direct effect on surrounding electronic equipment. Three different types of EMI can be found in the world. This includes:

1. Inherent EMI is located within an electrical device. It is noise that is created through thermal agitation (electrons moving through circuit resistor).
2. Natural EMI is caused by natural events that require no help from humans. EMI is found during snow and electrical storms. It is found in rain particles, as well as solar radiation. Many call this sort of interference atmospheric noise. Natural EMI will cause issues during RF communication in older equipment.

	Modern equipment does not suffer from natural EMI as much as it does from human-made EMI. 3. Lastly, electronic equipment produces human-made EMI. Most electronics are guilty of generating EMI, but this is found most commonly in transmitters, power lines, generators, electrical collectors, and igniters. The electronic equipment mentioned above has been known to cause extremely high levels of EMI. This will degrade the operations of the particular device and all other devices it comes in contact with.
10	**Impedence Matching** It's an essential requirement for transfer of power from line to load. A transmission line transfers max power when line is terminated with its characteristic impedence. Condition for impedence matching is that load impedence must be equal to characteristic impedence

10	

ie; $Z_R = Z_0$

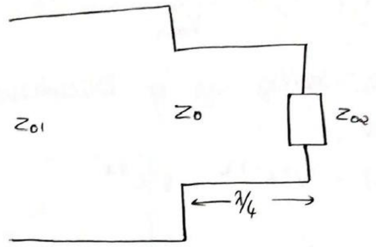
The transmission line satisfying this condition is called matched line. For such a line magnitude of $|\Gamma_R| = 0$, $S = 1$ and there is no reflection at the load end. As a result reflected wave is absent. The line carries forward travelling wave. The line impedance is constant and is equal to characteristic impedence Z_0.

Practical lines will have impedence mismatch ie; $Z_R \neq Z_0$. The matching can be provided by two techniques.

(i) quarter wave transformer
(ii) stub matching

Quarter Wave Transformer

Quarter wave transformer can be used for connecting two transmission lines having different characteristic impedence.



Let Z_{01} & Z_{02} are the two characteristic impedances of two lines which are to be matched, then Z_0 = geometric mean of characteristic impedence Z_{01} & Z_{02}

$$Z_0 = \sqrt{Z_{01} Z_{02}}$$

Disadvantage

i) Sensitive to the changes in frequency

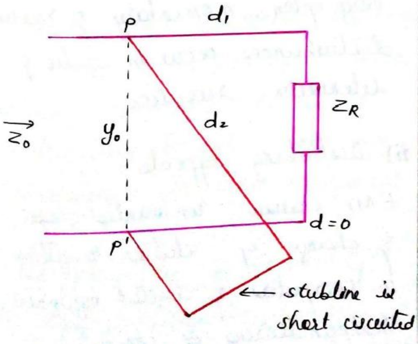
ii) For narrow band of frequencies, a quarter wave transformer works with tolerable change in VSWR.

Stub matching

Suitable sections of short circuit lines are called stubs connected in parallel with line at selective positions to produce impedence matching. Since stub line is reactive it will not absorb any power. This method of impedence matching of a line is called stub matching.

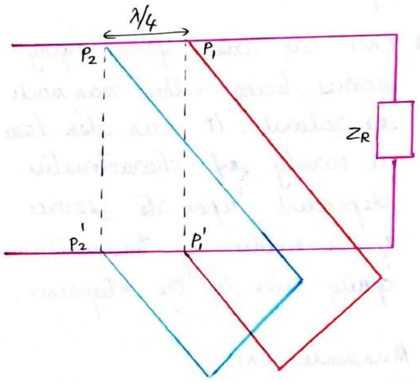
If a single stub line is used it is called single stub matching. If two stub lines are used it is called double stub matching.

Single stub matching



For impedence matching a single stub of definite length should be connected to the line at a suitable location. Since the stub cannot be placed physically in ideal locations, single stub matching is impractical.

Double stub matching



Double stub matching is flexible so that they can be connected to the line at any arbitrary location. Hence the separation b/w the stubs should be $\lambda/4$. It can be easily tuned by adjusting positions of stub. It gives a greater range of

PART B

(Answer one full question from each module, each question carries 14 marks)

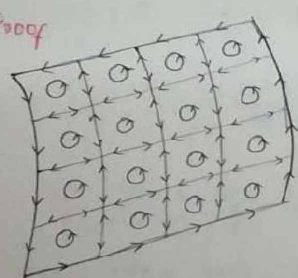
11a		

Stoke's Theorem

Stokes theorem states that the circulation of a vector field $\vec{A}$ around a closed path L is equal to the surface integral of the curl of $\vec{A}$ over the open surface s bounded by L provided that $\vec{A}$ and $\nabla \times \vec{A}$ are continous on s.

$$\iint_S (\nabla \times \vec{A}) \cdot \vec{ds} = \oint_L \vec{A} \cdot \vec{dl}$$

Proof



Consider the surface s is subdivided into a large number of cells with areas $ds_1, ds_2 \ldots$ If the k^{th} cell has surface area Δs_k & is bounded by path L_k,

R.H.S =

$$\oint_L \vec{A} \cdot \vec{dl} = \sum_k \oint_{L_k} \vec{A} \cdot \vec{dl}$$

$$= \sum_k \frac{\oint_{L_k} \vec{A} \cdot \vec{dl}}{\Delta S_k} \Delta S_k$$

$$= \sum_k \lim_{\Delta S_k \to 0} \frac{\oint_{L_k} \vec{A} \cdot \vec{dl}}{\Delta S_k} \Delta S_k$$

$$= \sum_k (\nabla \times \vec{A}) \Delta S_k$$

$$= \iint_S (\nabla \times \vec{A}) \cdot \vec{ds}$$

= L.H.S

Hence the proof

* Evaluate both sides of the stoke's theorem for the vector field $\vec{H} = 6xy\,\hat{a}_x - 3y^2\,\hat{a}_y$. The rectangular path along the region is $2 \le x \le 5$,

$-1 \le y \le 1$,

$z = 0$

11b		(worked answer below)

$$\vec{P} = xy\hat{a}_x + y^2\hat{a}_y + xz\hat{a}_z$$

$$\nabla\times\vec{P} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P_x & P_y & P_z \end{vmatrix} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & y^2 & xz \end{vmatrix}$$

$$= \hat{a}_x(0-0) - \hat{a}_y(z-0) + \hat{a}_z(0-x)$$

$$= -z\hat{a}_y - x\hat{a}_z$$

$$\vec{Q} = \rho z^2\hat{a}_\rho + \rho\sin^2\phi\,\hat{a}_\phi + 2\rho z\sin^2\phi\,\hat{a}_z$$

$$\nabla\times\vec{Q} = \frac{1}{\rho}\begin{vmatrix} \hat{a}_\rho & \rho\hat{a}_\phi & \hat{a}_z \\ \frac{\partial}{\partial\rho} & \frac{\partial}{\partial\phi} & \frac{\partial}{\partial z} \\ Q_\rho & \rho Q_\phi & Q_z \end{vmatrix}$$

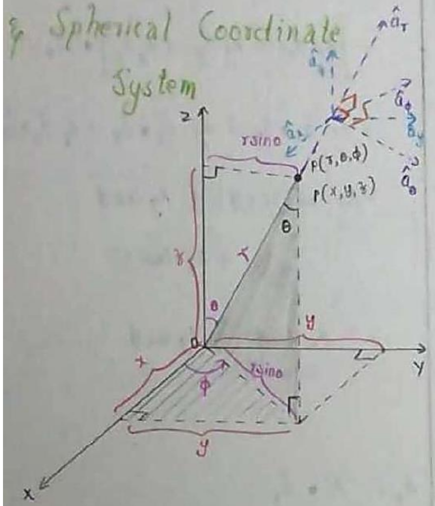
$$= \frac{1}{\rho}\begin{vmatrix} \hat{a}_\rho & \rho\hat{a}_\phi & \hat{a}_z \\ \frac{\partial}{\partial\rho} & \frac{\partial}{\partial\phi} & \frac{\partial}{\partial z} \\ \rho z^2 & \rho\sin^2\phi & 2\rho z\sin^2\phi \end{vmatrix}$$

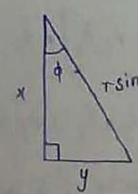
$$\nabla\times\vec{Q} = \frac{1}{\rho}\left[\hat{a}_\rho(2\rho z\sin 2\phi) + \hat{a}_\phi\,\rho(2z\sin^2\phi - 2\rho z) + \hat{a}_z(\sin^2\phi)\right]$$

$$= 2z\sin 2\phi\,\hat{a}_\rho - 2z(\sin^2\phi - \rho)\,\hat{a}_\phi + \sin^2\phi\,\hat{a}_z$$

12a

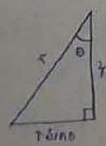
Relation between Cartesian & Spherical Coordinate System





$$\cos\phi = \frac{x}{r\sin\theta}$$

$$\sin\phi = \frac{y}{r\sin\theta}$$



$$\cos\theta = \frac{z}{r}$$

$$z = r\cos\theta$$

$$x = r\sin\theta\cos\phi$$
$$y = r\sin\theta\sin\phi$$
$$z = r\cos\theta$$

$$x^2+y^2+z^2 = r^2\sin^2\theta\cos^2\phi + r^2\sin^2\theta\sin^2\phi + r^2\cos^2\theta$$

$$= r^2\sin^2\theta\left[\cos^2\phi + \sin^2\phi\right] + r^2\cos^2\theta$$

$$= r^2\sin^2\theta + r^2\cos^2\theta$$

$$= r^2\left[\sin^2\theta + \cos^2\theta\right]$$

$$= r^2$$

$$r = \sqrt{x^2+y^2+z^2}$$

$$\cos\theta = \frac{z}{r}$$

$$\theta = \cos^{-1}\frac{z}{r}$$

$$\theta = \cos^{-1}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)$$

$$\tan\phi = \frac{y}{x}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$r = \sqrt{x^2+y^2+z^2}$$
$$\theta = \cos^{-1}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)$$
$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

12b

(12b) $$\vec{A} = \rho \sin\phi \, \hat{a}_\rho + \rho\cos\phi \, \hat{a}_\phi - 2z\hat{a}_z$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

$$= \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \rho\sin\phi \\ \rho\cos\phi \\ -2z \end{bmatrix}$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \rho\cos\phi\sin\phi - \rho\sin\phi\cos\phi \\ \rho\sin^2\phi + \rho\cos^2\phi \\ -2z \end{bmatrix} = \begin{bmatrix} 0 \\ \rho \\ -2z \end{bmatrix}$$

$$\vec{A} = \rho\hat{a}_y - 2z\hat{a}_z$$

$$= \sqrt{x^2+y^2}\,\hat{a}_y - 2z\hat{a}_z$$

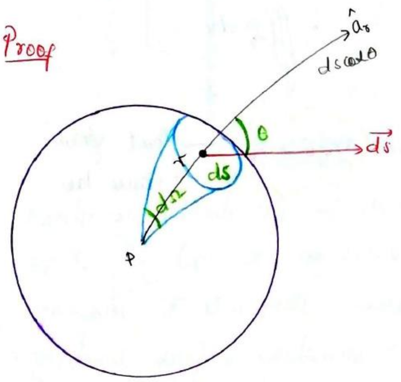
13a

Gauss Law

- Gauss law states that the total electric flux φ through any closed surface is equal to the total charge enclosed by that surface

$$\oint_S \vec{D} \cdot \vec{ds} = Q$$ → Integral form of Gauss law

Proof



Consider a small surface ds which is a part of a solid sphere. It makes a solid angle $d\Omega$ at P. r is the line joining the centre of the area ds to the point P. The direction of $\vec{ds}$ is given by the unit vector normal to ds in the outward direction.

Solid angle $d\Omega$ = radial component of surface area ds / The square of the distance b/w the surface ds & point P

$$d\Omega = \frac{ds\cos\theta}{r^2}$$

For a solid sphere total solid angle is given by

$$\Omega = \oint d\Omega$$

$$\Omega = \int d\Omega$$

$$\Omega = \oint_S \frac{ds\cos\theta}{r^2}$$

For solid sphere $\theta = 0$

$$\Omega = \frac{1}{r^2}\oint_S ds$$

$$\Omega = \frac{1}{r^2} 4\pi r^2$$

$$\Omega = 4\pi$$

The electric flux through ds is

$$d\psi = \vec{D} \cdot \vec{ds}$$

$$= D \cdot ds \cdot \cos\theta$$

$$D = \frac{Q}{4\pi r^2}$$ Q → Total charge, $4\pi r^2$ → Total surface area

$$d\psi = \frac{Q}{4\pi r^2} \cdot ds \cdot \cos\theta$$

Total electric flux,

$$\psi = \int d\psi$$

$$\psi = \frac{Q}{4\pi} \oint_S \frac{ds \cos\theta}{r^2}$$

$$\psi = \frac{Q}{4\pi} \times \Omega$$

$$\psi = \frac{Q}{4\pi} \times 4\pi$$

$$\underline{\underline{\psi = Q}}$$

Hence the proof.

$\therefore$ Total flux through a closed surface

$$\oint_S \vec{D} \cdot \vec{ds} = Q$$

Gauss divergence theorem

From Gauss law,

$$\oint_S \vec{D} \cdot \vec{ds} = Q \quad \text{------} \ (1)$$

If ρ_v is the volume charge density then the total charge Q enclosed by the volume V is given by

$$Q = \iiint_V \rho_v \, dv \quad \text{----} \ (2).$$

Equate (1) & (2),

$$\oint_S \vec{D} \cdot \vec{ds} = \iiint_V \rho_v \, dv$$

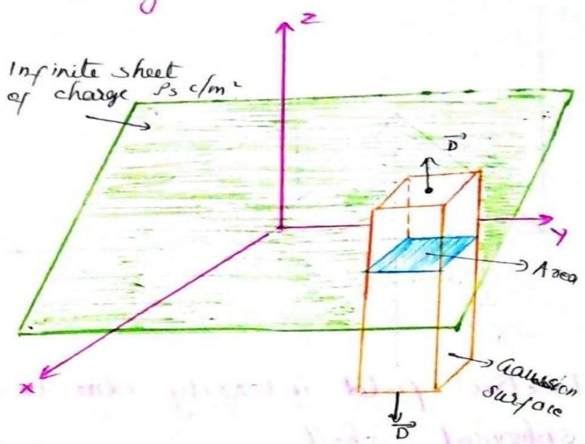
Using divergence theorem $\oint_S$ can be converted to $\iiint_V$

$$\therefore \oint_S \vec{D} \cdot \vec{ds} = \iiint_V \nabla \cdot \vec{D} \, dv$$

$$= \iiint_V \rho_v \, dv$$

$$\Longrightarrow \boxed{\nabla \cdot \vec{D} = \rho_v} \longrightarrow \text{Point form of Gauss law}$$

iv) Electric field intensity due to infinite sheet of charge



Consider a Gaussian surface, a rectangular parallelopipe at $z=0$. Two of its faces are parallel to the sheet.

According to Gauss law,

$$Q = \oint_S \vec{D} \cdot \vec{ds} = D\left[\iint_{top} ds + \iint_{bottom} ds\right]$$

$D.ds.\cos\theta$
$= D ds \quad \theta = 0$

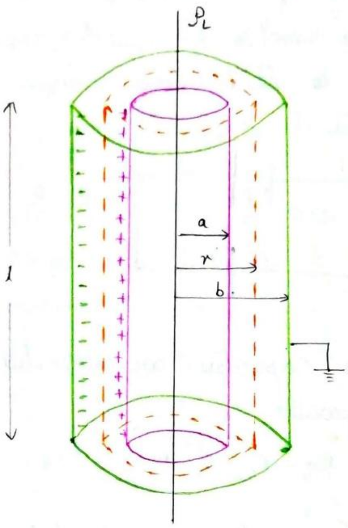
$\vec{D}$ has no components along $\hat{a}_x$ & $\hat{a}_y$. Direction of flux is always normal to the sheet ie; $\hat{a}_z$

If the top & bottom of the box has area A, then

$$Q = D(A + A)$$

$$Q = 2DA \quad \text{--- (1)}$$

		$Q = \rho_s A$ ---(2) area on the sheet (blue) Equate (1) & (2) $\rho_s \cancel{A} = 2D\cancel{A}$ $\vec{D} = \frac{\rho_s}{2} \hat{a}_z$ $\vec{E} = \vec{D}/\varepsilon$ $\vec{E} = \frac{\rho_s}{2\varepsilon} \hat{a}_z$

14a	**Capacitance of Co-axial Cable**  Consider a coaxial cable with radii a & b. The space b/w the cylinder is filled with a dielectric of relative permittivity ϵ_r. Let ρ_L be the line charge density. A charge of +Q is given to the inner conductor. This induces a charge ~~+Q & -Q~~ -Q & +Q on the inner & outer surface of the outer conductor. The outer surface of the outer conductor is earthed which neutralise the +Q charge on its outer surface.

These charges +Q & -Q produce an electric field b/w the two conductors which are radially outward.

Consider a Gaussian surface of radius r & length l.

By Gauss law,

$$Q = 2\pi r l \vec{D}$$

$$\vec{D} = \frac{Q}{2\pi r l} \hat{a}_r$$

$$Q = \rho_L l$$

$$\vec{D} = \frac{\rho_L l}{2\pi r l} \hat{a}_r$$

$$\vec{D} = \frac{\rho_L}{2\pi r} \hat{a}_r$$

$$\vec{E} = \frac{\vec{D}}{\varepsilon}$$

$$\vec{E} = \frac{\rho_L}{2\pi \varepsilon r} \hat{a}_r$$

$$dV = -E \cdot dr$$

Point at potential P, $V = \int_b^a - E \cdot dr$

$$V = -\int_b^a \frac{\rho_L}{2\pi\varepsilon r} dr$$

$$V = -\frac{\rho_L}{2\pi\varepsilon} \int_b^a \frac{dr}{r}$$

$$V = -\frac{\rho_L}{2\pi\varepsilon} \left[\ln r\right]_b^a$$

$$V = -\frac{\rho_L}{2\pi\varepsilon} \left[\ln a - \ln b\right]$$

$$V = \frac{\rho_L}{2\pi\varepsilon} \left[\ln b - \ln a\right]$$

$$V = \frac{\rho_L}{2\pi\varepsilon} \ln(b/a)$$

$$C = \frac{Q}{V} = \frac{\cancel{\rho_L} l}{\frac{\cancel{\rho_L}}{2\pi\varepsilon} \ln(b/a)}$$

$$C = \frac{2\pi\varepsilon l}{\ln(b/a)}$$

Capacitance per length $= \dfrac{2\pi\varepsilon}{\ln(b/a)}$ F/m

14b		

Poisson's & Laplace's Equations

Poisson's & Laplace eqns can be easily derived from point form of Gauss law, ie; $\nabla \cdot \vec{D} = \rho_V$

$$\nabla \cdot \vec{D} = \rho_V$$

$$\nabla \cdot \varepsilon \vec{E} = \rho_V$$

$$\varepsilon \left[\nabla \cdot \vec{E} \right] = \rho_V$$

$$\nabla \cdot \vec{E} = \frac{\rho_V}{\varepsilon}$$

$$\nabla \cdot (-\nabla V) = \frac{\rho_V}{\varepsilon}$$

$$\nabla \cdot \nabla V = -\frac{\rho_V}{\varepsilon}$$

$$\nabla^2 V = -\frac{\rho_V}{\varepsilon} \longrightarrow \text{Poisson's eqn}$$

In a charge free region $\rho_V = 0$

$$\nabla^2 V = 0 \longrightarrow \text{Laplace eqn}$$

Cartesian System

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} = -\frac{\rho_V}{\varepsilon}$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

Cylindrical Coordinate System

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho}\left(\rho \frac{\partial V}{\partial \rho}\right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho_V}{\varepsilon}$$

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho}\left(\rho \frac{\partial V}{\partial \rho}\right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

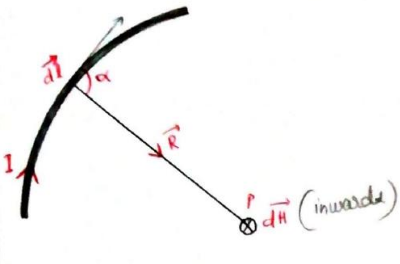
Spherical Coordinate System

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta} \frac{\partial^2 V}{\partial \phi^2} = -\frac{\rho_V}{\varepsilon}$$

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta} \frac{\partial^2 V}{\partial \phi^2} = 0$$

15a

Biot - Savart Law



Biot - Savart's law states that the magnetic field intensity dH produced at a point P, by the differential current element Idl is proportional to the product Idl and the sine of the angle α between the element & the line joining P to the element and is inversely proportional to the square of the distance R between P and the element.

ie; $dH \propto \frac{Idl \sin\alpha}{R^2}$

$$dH = \frac{K\, Idl \sin\alpha}{R^2}$$

where $K = \frac{1}{4\pi}$

$$dH = \frac{I\, dl \sin\alpha}{4\pi R^2}$$

$\hat{a}_R$ is the unit vector in the direction from the differential current element to the point P.

$$\hat{a}_R = \frac{\vec{R}}{|\vec{R}|}$$

$$\vec{dl} \times \hat{a}_R = |\vec{dl}|\,|\hat{a}_R| \sin\alpha$$

$$= dl \sin\alpha$$

$$\vec{dH} = \frac{I(\vec{dl} \times \hat{a}_R)}{4\pi R^2}$$

or

$$\vec{dH} = \frac{I(\vec{dl} \times \vec{R})}{4\pi R^3}$$

We can use right handed screw rule to determine the direction of $\vec{dH}$.

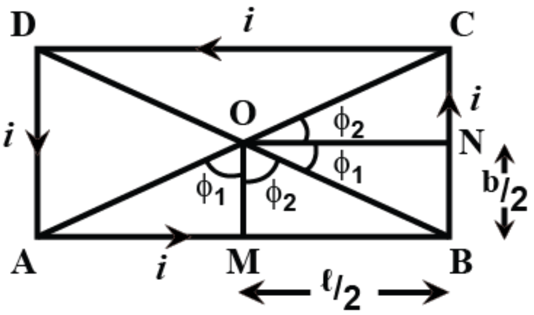
Total magnetic field intensity

$$\vec{H} = \oint \vec{dH}$$

ie; $\vec{H} = \oint \frac{I(\vec{dl} \times \hat{a}_R)}{4\pi R^2}$ or

$$\vec{H} = \oint \frac{I(\vec{dl} \times \vec{R})}{4\pi R^3}$$

15b



The magnetic fields produced at the center of rectangle by all four sides of rectangle will be in same direction, therefore, total magnetic field at center O,

$$B = B_{AB} + B_{BC} + B_{CD} + B_{DA}$$

The magnetic field at a point, due to a straight current carrying conductor is given by,

$$B = \frac{\mu_0}{4\pi}.\frac{i}{d}(\sin\phi_1 + \sin\phi_2)$$

For length AB and CD,

$$d = b/2$$

and $\sin\phi_1 = \sin\phi_2 = AM/AO = \dfrac{l/2}{(\sqrt{l^2+b^2})/2} = \dfrac{l}{\sqrt{l^2+b^2}}$

Hence,

$$B_{AB} = B_{CD} = \frac{\mu_0}{4\pi}.\frac{i}{b/2}\left(\frac{l}{\sqrt{l^2+b^2}} + \frac{l}{\sqrt{l^2+b^2}}\right)$$

or $B_{AB} = B_{CD} = \dfrac{\mu_0}{\pi}.\dfrac{i}{b}\left(\dfrac{l}{\sqrt{l^2+b^2}}\right)$ eq1

For breadth BC and DA,

$$d = l/2$$

and $\sin\phi_1 = \sin\phi_2 = BN/BO = \dfrac{b/2}{(\sqrt{l^2+b^2})/2} = \dfrac{b}{\sqrt{l^2+b^2}}$

Hence,

$$B_{BC} = B_{DA} = \frac{\mu_0}{4\pi}.\frac{i}{l/2}\left(\frac{b}{\sqrt{l^2+b^2}} + \frac{b}{\sqrt{l^2+b^2}}\right)$$

or $B_{BC} = B_{DA} = \dfrac{\mu_0}{\pi}.\dfrac{i}{l}\left(\dfrac{b}{\sqrt{l^2+b^2}}\right)$ eq2

Hence,

$$B = \frac{\mu_0}{\pi}.\frac{i}{b}\left(\frac{l}{\sqrt{l^2+b^2}}\right) + \frac{\mu_0}{\pi}.\frac{i}{l}\left(\frac{b}{\sqrt{l^2+b^2}}\right) + \frac{\mu_0}{\pi}.\frac{i}{b}\left(\frac{l}{\sqrt{l^2+b^2}}\right)$$

$$+\frac{\mu_0}{\pi}.\frac{i}{l}\left(\frac{b}{\sqrt{l^2+b^2}}\right)$$

or $B = \dfrac{2\mu_0 i}{\pi}.\dfrac{\sqrt{l^2+b^2}}{lb}$

16a		Boundary Conditions for Electric Field

Boundary Conditions for Electric Field

- The conditions existing at the boundary of the two media when the field passes from one media to other are called as boundary conditions.

- For boundary conditions, following relations are required

$$\oint_S \vec{D} \cdot \vec{ds} = Q$$

$$\oint_L \vec{E} \cdot \vec{dl} = 0$$

$$\vec{E} = \vec{E}_{tan} + \vec{E}_N$$

$$\vec{D} = \vec{D}_{tan} + \vec{D}_N$$

tangential to the boundary

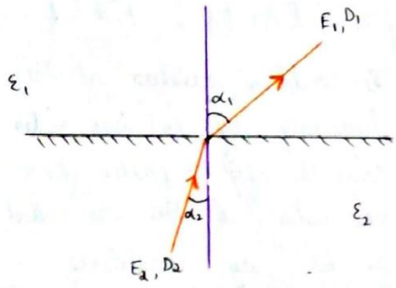
Normal to the boundary

i). Dielectric Boundary Conditions

Consider the boundary b/w two dielectrics with different properties. Let the two media have the permittivities

$$\varepsilon_1 = \varepsilon_0 \varepsilon_{r1} \quad \& \quad \varepsilon_2 = \varepsilon_0 \varepsilon_{r2}$$

The lines of $\vec{E}$ & $\vec{D}$ will be refracted when passing from media 2 to media 1.



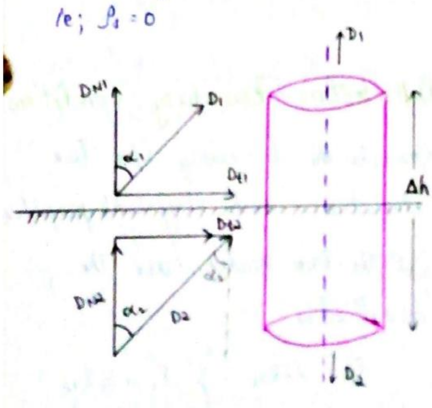
Let D_1 & D_2 be the flux densities and E_1 & E_2 be the field intensities in respective media.

Then $D_1 = \varepsilon E_1 \implies D_1 = \varepsilon_0 \varepsilon_{r1} E_1$

$D_2 = \varepsilon E_2 \implies D_2 = \varepsilon_0 \varepsilon_{r2} E_2$

Assume that there is no free charge on the interface b/w the dielectrics

ie; $\rho_s = 0$



Consider a pill box with very small thickness enclosed in a portion of boundary.

From Gauss law, the $\oint_S \vec{D} \cdot \vec{ds} = Q_{enc}$

but here $\rho_s = 0$

$$\therefore \int_{top} \vec{D} \cdot \vec{ds} + \int_{bottom} \vec{D} \cdot \vec{ds} + \int_{lateral} \vec{D} \cdot \vec{ds} = 0$$

But $\Delta h \longrightarrow 0$, so $\int_{lateral} \vec{D} \cdot \vec{ds} = 0$

Also, $\int_{top} \vec{D} \cdot \vec{ds} = D_{N1}\, ds$.

$$\int_{bottom} \vec{D} \cdot \vec{ds} = -D_{N2}\, ds$$

$$\therefore \int_{top} \vec{D} \cdot \vec{ds} + \int_{bottom} \vec{D} \cdot \vec{ds} = 0$$

$$D_{N1}\, ds - D_{N2}\, ds = 0$$

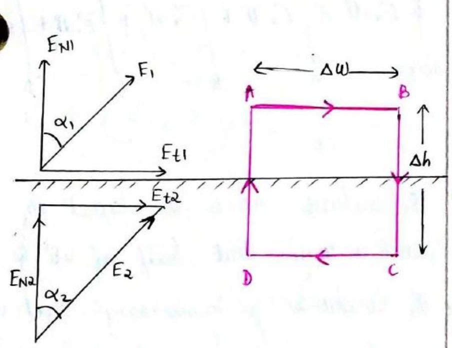
$$\implies D_{N1}\, ds = D_{N2}\, ds$$

ie; $D_{N1} = D_{N2}$ ....(1)

ie; the normal component of flux density $\vec{D}$ is continuous at the boundary b/w two perfect dielectric as the normal displacement flux entering one face is same as that leaving the other face.

Consider the electric field intensity at the boundary. The potential around any closed path in an electric field must be zero.

ie; $V = \oint \vec{E} \cdot \vec{dl} = 0$



Consider the path ABCD.

$$\int_{ABCD} \vec{E} \cdot \vec{dl} = \int_A^B \vec{E} \cdot dl + \int_B^C \vec{E} \cdot \vec{dl} + \int_C^D \vec{E} \cdot \vec{dl} + \int_D^A \vec{E} \cdot \vec{dl} = 0$$

Since $\Delta h \to 0$ $\int_D^A \vec{E} \cdot \vec{dl} = \int_B^C \vec{E} \cdot \vec{dl} = 0$

$$\therefore \int_{ABCD} \vec{E} \cdot \vec{dl} = \int_A^B \vec{E} \cdot \vec{dl} + \int_C^D \vec{E} \cdot \vec{dl} = 0$$

$\Longrightarrow E_{t1} \Delta W - E_{t2} \Delta W = 0$

$E_{t1} \cancel{\Delta W} = E_{t2} \cancel{\Delta W}$

ie; $E_{t1} = E_{t2}$ ...... (2)

ie; The tangential components of fields E_1 & E_2 at the boundary are equal.

From (1), normal components of D are equal.

$D_1 \cos \alpha_1 = D_2 \cos \alpha_2$ ... (3)

From (2), tangential components of E are equal

$E_1 \sin \alpha_1 = E_2 \sin \alpha_2$ ... (4)

$$\frac{(4)}{(3)} = \frac{E_1 \sin \alpha_1}{D_1 \cos \alpha_1} = \frac{E_2 \sin \alpha_2}{D_2 \cos \alpha_2}$$

$$\frac{E_1}{D_1} \tan \alpha_1 = \frac{E_2}{D_2} \tan \alpha_2$$

$$\frac{\tan \alpha_1}{\tan \alpha_2} = \frac{E_2}{D_2} \frac{D_1}{E_1}$$

$$= \frac{E_2}{E_1} \frac{D_1}{D_2}$$

	$$\frac{\tan\alpha_1}{\tan\alpha_2} = \frac{\cancel{E_2}}{\cancel{E_1}} \frac{\cancel{\varepsilon_0}\varepsilon_{r1}\cancel{E_1}}{\cancel{\varepsilon_0}\varepsilon_{r2}\cancel{E_2}}$$ $$\frac{\tan\alpha_1}{\tan\alpha_2} = \frac{\varepsilon_{r1}}{\varepsilon_{r2}} = \frac{\varepsilon_1}{\varepsilon_2}$$ Law of refraction ∴ Boundary conditions b/w two dielectrics are; 1. $D_{N1} = D_{N2}$ 2. $E_{t1} = E_{t2}$ 3. $\frac{\tan\alpha_1}{\tan\alpha_2} = \frac{\varepsilon_{r1}}{\varepsilon_{r2}}$

16b	## Continuity Equation Continuity eqn is based on law of conservation of charges. It states that "charges can neither be created nor be destroyed". Consider a closed surface 's' with current density vector J. Then the total current I crossing the surface is, $$I = \oint_s \vec{J} \cdot \vec{ds}$$ According to conservation of charge there must be decrease of an equal amount of +ve charges inside the closed surface. Hence the outward rate of flow of +ve charge get balanced by the rate of decrease of charge inside the closed surface. ie; $$I = \oint_s \vec{J} \cdot \vec{ds} = -\frac{dQ}{dt} \quad \text{....①}$$ where Q is the charge inside the closed surface 's'. -ve sign indicates decrease in charge.

If the current is entering

$$-I = \oint_S \vec{J} \cdot \vec{ds} = +\frac{dQ}{dt} \quad \ldots ②$$

Eqns ① & ② are known as integral form of continuity eqn.

$$\oint_S \vec{J} \cdot \vec{ds} = \iiint_V (\nabla \cdot \vec{J})\, dv$$

(using divergence theorem)

Also $Q = \iiint_V \rho_v \, dv$

since $I = \oint_S \vec{J} \cdot \vec{ds} = -\frac{dQ}{dt}$

$$\therefore \iiint_V (\nabla \cdot \vec{J})\, dv = -\frac{d}{dt}\left[\iiint_V \rho_v \, dv\right]$$

$$= \iiint_V -\frac{d}{dt} \rho_v \, dv$$

For a constant surface the derivative become partial derivative

$$\therefore \iiint_V (\nabla \cdot \vec{J})\, dv = \iiint_V -\frac{\partial \rho_v}{\partial t} \, dv$$

ie; $\nabla \cdot \vec{J} = -\frac{\partial \rho_v}{\partial t}$ → Point form/ diff form of continuity eqn

Continuity eqn states that current or charge per second diverging from ~~a small volume~~ per unit volume is equal to the time rate of decrease of charge per unit volume at every point.

For steady current $\frac{\partial \rho_v}{\partial t} = 0$

$$\Longrightarrow \nabla \cdot \vec{J} = 0$$

Maxwell's Equation

i) From Modified form of Ampere's Circuital Law

From Ampere Circuital law,

$$\oint_L H \cdot \vec{dl} = I_{enc} = \oiint_S \vec{J} \cdot \vec{ds} \quad \text{(Integral form)}$$

Also $\nabla \times \vec{H} = \vec{J} \quad \ldots ①$ (Point form)

Taking divergence on both sides of eqn ①,

$$\nabla \cdot (\nabla \times \vec{H}) = \nabla \cdot \vec{J}$$

Since divergence of any curl is zero $\nabla \cdot (\nabla \times \vec{H})$ must be zero.

17a		**Wave Equation from Maxwell's Equation** Electromagnetic wave eqn can be explained with the help of Maxwell's eqns. $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \left\{ \vec{B} = \mu \vec{H} \right.$ $\nabla \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t} \quad \cdots\cdots (1)$ $\nabla \times \vec{H} = \vec{J_c} + \frac{\partial \vec{D}}{\partial t} \quad \left\{ \begin{array}{l} \vec{J_c} = \sigma \vec{E} \\ \vec{D} = \varepsilon \vec{E} \end{array} \right.$ $\nabla \times \vec{H} = \sigma \vec{E} + \varepsilon \frac{\partial \vec{E}}{\partial t} \cdots (2)$

$\nabla \cdot \vec{B} = \nabla \cdot \vec{H} = 0$ ---- ③

$\nabla \cdot \vec{D} = \nabla \cdot \vec{E} = 0$ ---- ④

$\left\{ \nabla \cdot \vec{D} = \rho_V = 0 \implies \nabla \cdot \vec{E} = 0 \right\}$

Apply curl operation on both sides of eqn ①

$$\nabla \times (\nabla \times \vec{E}) = \nabla \times \left(-\mu \frac{\partial \vec{H}}{\partial t}\right)$$

$$\nabla \times (\nabla \times \vec{E}) = -\mu \frac{\partial}{\partial t}(\nabla \times \vec{H}) \text{ --- ⑤}$$

Substitute ② in ⑤

$$\nabla \times (\nabla \times \vec{E}) = -\mu \frac{\partial}{\partial t}\left(\sigma \vec{E} + \varepsilon \frac{\partial \vec{E}}{\partial t}\right)$$

$$\nabla \times (\nabla \times \vec{E}) = -\mu\sigma \frac{\partial \vec{E}}{\partial t} - \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2} \text{ -- ⑥}$$

By vector identity,

$$\nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E}$$

We know $\nabla \cdot \vec{E} = 0$ from eqn ④

$\implies \nabla \times (\nabla \times \vec{E}) = -\nabla^2 \vec{E}$

Then eqn ⑥ $\implies$

$$-\nabla^2 \vec{E} = -\mu\sigma \frac{\partial \vec{E}}{\partial t} - \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{E} = \mu\sigma \frac{\partial \vec{E}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2} \text{ --- ⑦}$$

Now multiply both sides of eqn by ε.

$$\nabla^2 (\varepsilon \vec{E}) = \mu\sigma \frac{\partial (\varepsilon \vec{E})}{\partial t} + \mu\varepsilon \frac{\partial^2 (\varepsilon \vec{E})}{\partial t^2}$$

$$\nabla^2 \vec{D} = \mu\sigma \frac{\partial \vec{D}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{D}}{\partial t^2} \text{ ...}$$

~~7 & 7* are the second~~

Eqn ⑦ is the II order partial differential eqn of $\vec{E}$ and $\vec{E}$ must obey this eqn.

$\therefore \vec{E}$ is a wave.

Apply curl on both sides of eqn ②

$$\nabla \times (\nabla \times \vec{H}) = \nabla \times \left(\sigma \vec{E} + \varepsilon \frac{\partial \vec{E}}{\partial t}\right)$$

$$\nabla \times (\nabla \times \vec{H}) = \sigma (\nabla \times \vec{E}) + \varepsilon \frac{\partial}{\partial t}(\nabla \times \vec{E}) \text{ ...⑧}$$

Substitute eqn ① in ⑧

$$\nabla \times (\nabla \times \vec{H}) = \sigma\left(-\mu \frac{\partial \vec{H}}{\partial t}\right) + \varepsilon \frac{\partial}{\partial t}\left(-\mu \frac{\partial \vec{H}}{\partial t}\right)$$

$$\nabla \times (\nabla \times \vec{H}) = -\mu\sigma \frac{\partial \vec{H}}{\partial t} - \mu\varepsilon \frac{\partial^2 \vec{H}}{\partial t^2} \quad \text{--- (9)}$$

By vector identity,

$$\nabla \times (\nabla \times \vec{H}) = \nabla (\nabla \cdot \vec{H}) - \nabla^2 \vec{H}$$

From eqn (3), $\nabla \cdot \vec{H} = 0$

Then, $\nabla \times (\nabla \times \vec{H}) = -\nabla^2 \vec{H}$

eqn (9) $\Longrightarrow$

$$-\nabla^2 \vec{H} = -\mu\sigma \frac{\partial \vec{H}}{\partial t} - \mu\varepsilon \frac{\partial^2 \vec{H}}{\partial t^2}$$

$$\nabla^2 \vec{H} = \mu\sigma \frac{\partial \vec{H}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{H}}{\partial t^2} \quad \text{---- (10)}$$

Now multiply both the sides with μ

$$\cancel{\nabla^2(\mu \vec{H}) = \mu\sigma \frac{\partial \vec{B}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{B}}{\partial t^2}}$$

$$\nabla^2(\mu \vec{H}) = \mu\sigma \frac{\partial(\mu \vec{H})}{\partial t} + \mu\varepsilon \frac{\partial^2(\mu \vec{H})}{\partial t^2}$$

$$\nabla^2 \vec{B} = \mu\sigma \frac{\partial \vec{B}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{B}}{\partial t^2}$$

Eqn (10) is the second order partial differential eqn of $\vec{H}$ and $\vec{H}$ must obey the eqn.

$\therefore \vec{H}$ is a wave

Wave Equation in Phasor Form

$$\nabla^2 \vec{E} = \mu\sigma \frac{\partial \vec{E}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{-- (1)}$$

$$\nabla^2 \vec{H} = \mu\sigma \frac{\partial \vec{H}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{H}}{\partial t^2} \quad \text{--- (2)}$$

Put $\frac{\partial}{\partial t} = j\omega$ in eqn (1)

$$\nabla^2 \vec{E} = \mu\sigma j\omega \vec{E} + \mu\varepsilon (j\omega)^2 \vec{E}$$

$$= j\mu\omega\sigma \vec{E} + j^2 \mu\varepsilon\omega^2 \vec{E}$$

$$= j\mu\omega (\sigma + j\varepsilon\omega) \vec{E}$$

$$\nabla^2 \vec{E} = \gamma^2 \vec{E}$$

where $\gamma = \sqrt{j\mu\omega(\sigma + j\varepsilon\omega)}$

		Put $\frac{\partial}{\partial t} = j\omega$ in eqn ② $\nabla^2 \vec{H} = \mu\sigma j\omega \vec{H} + \mu\varepsilon (j\omega)^2 \vec{H}$ $= j\mu\sigma\omega \vec{H} + j^2 \mu\varepsilon\omega^2 \vec{H}$ $= j\mu\omega(\sigma + j\varepsilon\omega)\vec{H}$ $\nabla^2 \vec{H} = \gamma^2 \vec{H}$ where $\gamma = \sqrt{j\mu\omega(\sigma + j\varepsilon\omega)}$

17b		

Plane Wave Propagation in Lossy dielectric Medium

$\sigma \neq 0, \ \varepsilon = \varepsilon_r \varepsilon_0, \ \mu = \mu_r \mu_0$

I. Wave eqns are,

$$\nabla^2 \vec{E} = \gamma^2 \vec{E}$$

$$\nabla^2 \vec{H} = \gamma^2 \vec{H}$$

Solutions are

$$E_x(z,t) = E_m^+ e^{-\alpha z} \cos(\omega t - \beta z) + E_m^- e^{\alpha z} \cos(\omega t + \beta z)$$

$$H_y(z,t) = H_m^+ e^{-\alpha z} \cos(\omega t - \beta z - \theta_n) + H_m^- e^{\alpha z} \cos(\omega t + \beta z - \theta_n)$$

II. Intrinsic Impedance

Modified Faraday's law,

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu \frac{\partial \vec{H}}{\partial t}$$

$$\begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = -\mu \frac{\partial}{\partial t}\left[H_x \hat{a}_x + H_y \hat{a}_y + H_z \hat{a}_z \right]$$

Assume $\vec{E}$ is in x direction, $\vec{H}$ is in y direction. Let the direction of propagation be in z direction.

$$E_y = E_z = 0$$

$$H_x = H_z = 0$$

$$\begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & 0 & 0 \end{vmatrix} = -\mu \frac{\partial}{\partial t}\left[0\hat{a}_x + H_y \hat{a}_y + 0\hat{a}_z \right]$$

$$\cancel{+\hat{a}_y}\left[+\frac{\partial E_x}{\partial z} \right] = -\mu \frac{\partial H_y}{\partial t} \cancel{\hat{a}_y}$$

$$\frac{\partial E_x}{\partial z} = -\mu \frac{\partial H_y}{\partial t} \quad \text{-----} (1)$$

Let the wave be in +ve z direction, then

$$E_x = E_m^+ e^{-\gamma z}$$

$$\frac{\partial E_x}{\partial z} = E_m^+ (-\gamma) e^{-\gamma z}$$

$$\frac{\partial E_x}{\partial z} = -\gamma E_m^+ e^{-\gamma z}$$

$$\frac{\partial E_x}{\partial z} = -\gamma E_x \quad \text{-----} (2)$$

Put ② in ①

$$-\gamma E_x = -\mu \frac{\partial H_y}{\partial t}$$

$$\gamma E_x = \mu \frac{\partial H_y}{\partial t}$$

Put $\frac{\partial}{\partial t} = j\omega$

$$\gamma E_x = \mu j\omega H_y$$

$$\eta = \frac{E_x}{H_y} = \frac{\mu j\omega}{\gamma} \quad \left\{ \gamma = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)} \right.$$

Intrinsic impedance $\eta = \frac{E_x}{H_y}$

$$\eta = \frac{j\omega\mu}{\sqrt{j\omega\mu(\sigma + j\omega\varepsilon)}}$$

$\frac{\sigma}{\omega\varepsilon}$ = tangent loss

$$\eta = \sqrt{\frac{(j\omega\mu)^{\cancel{2}}}{\cancel{j\omega\mu}(\sigma + j\omega\varepsilon)}}$$

$$\boxed{\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = |\eta| \angle \theta_n}$$

$$\boxed{\theta_n = \frac{1}{2}\left[90° - \tan^{-1}\left(\frac{\omega\varepsilon}{\sigma}\right)\right]}$$

θ_n : Phase angle b/w $\vec{E}$ & $\vec{H}$

low freq $\longrightarrow \theta_n \approx 45°$

high freq $\longrightarrow \theta_n = 0°$

III. Propagation Constant γ

$$\gamma = \alpha + j\beta \quad \text{---} \circledast$$

$$= \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)} \quad \text{---} ①$$

Squaring both sides

$$\alpha^2 - \beta^2 + 2j\alpha\beta = j\omega\mu(\sigma + j\omega\varepsilon)$$

$$\alpha^2 - \beta^2 + 2j\alpha\beta = -\omega^2\varepsilon\mu + j\omega\mu\sigma$$

Equating real & imaginary parts,

$$\alpha^2 - \beta^2 = -\omega^2\varepsilon\mu \quad \text{---} ②$$

$$2\alpha\beta = \omega\mu\sigma \quad \text{---} ③$$

$$\alpha = \frac{\omega\mu\sigma}{2\beta} \quad \text{---} ④$$

$$\beta = \frac{\omega\mu\sigma}{2\alpha} \quad ⑤$$

Put ④ in ②

$$\frac{\omega^2\mu^2\sigma^2}{4\beta^2} - \beta^2 = -\omega^2\varepsilon\mu$$

$$\omega^2\mu^2\sigma^2 - 4\beta^4 = -4\omega^2\beta^2\varepsilon\mu$$

$$4\beta^4 - 4\mu\varepsilon\omega^2\beta^2 - \omega^2\mu^2\sigma^2 = 0$$

Solving,

$$\beta^2 = \frac{4\omega^2\varepsilon\mu \pm \sqrt{16\omega^4\varepsilon^2\mu^2 - [4\times 4(-\omega^2\mu^2\sigma^2)]}}{2\times 4}$$

$$\beta^2 = \frac{4\omega^2\varepsilon\mu \pm \sqrt{16\omega^4\varepsilon^2\mu^2 + 16\omega^2\mu^2\sigma^2}}{8}$$

$$\beta^2 = \frac{4\omega^2\varepsilon\mu + \sqrt{16\omega^4\varepsilon^2\mu^2\left(1+\frac{\sigma^2}{\omega^2\varepsilon^2}\right)}}{8}$$

$$\beta^2 = \frac{4\omega^2\varepsilon\mu + 4\omega^2\varepsilon\mu\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}}{8}$$

$$\beta^2 = \frac{\omega^2\varepsilon\mu}{2} + \frac{\omega^2\varepsilon\mu}{2}\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}$$

$$\beta^2 = \frac{\omega^2\varepsilon\mu}{2}\left[1+\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}\right] \quad \text{---} (6)$$

$$\beta = \omega\sqrt{\frac{\varepsilon\mu}{2}\left[1+\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}\right]}$$

Put (6) in (2)

$$\alpha^2 - \frac{\omega^2\varepsilon\mu}{2}\left[1+\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}\right] = -\omega^2\varepsilon\mu$$

$$\alpha^2 - \frac{\omega^2\varepsilon\mu}{2} - \frac{\omega^2\varepsilon\mu}{2}\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} = -\omega^2\varepsilon\mu$$

$$\alpha^2 = -\omega^2\varepsilon\mu + \frac{\omega^2\varepsilon\mu}{2} + \frac{\omega^2\varepsilon\mu}{2}\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}$$

$$\alpha^2 = -\frac{\omega^2\varepsilon\mu}{2} + \frac{\omega^2\varepsilon\mu}{2}\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}$$

$$\alpha^2 = \frac{\omega^2\varepsilon\mu}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1\right]$$

$$\alpha = \omega\sqrt{\frac{\varepsilon\mu}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1\right]}$$

IV. Phase Velocity (Propagation Velocity) V_p

It's the velocity with which the phase of the wave propagates in space.

Consider a +ve $\vec{E}$ wave travelling in x direction

$E_x(z,t) = E_m^+ \cos(\omega t - \beta z)$

For constant phase, $(\omega t - \beta z)$ is a constant

Differentiating w.r.t 't',

$$\omega - \beta \frac{dz}{dt} = 0$$

$$\frac{dz}{dt} = \frac{\omega}{\beta}$$

Phase velocity, $V_p = \frac{dz}{dt} = \frac{\omega}{\beta}$

$$\beta = \frac{2\pi}{\lambda}$$

Then $V_p = \frac{\omega \lambda}{2\pi}$

$$\omega = 2\pi f$$

$$V_p = \frac{\cancel{2\pi} f \lambda}{\cancel{2\pi}}$$

$$V_p = f\lambda$$

V. Group Velocity, V_g

It's the velocity with which the overall envelope shape of the wave's amplitude - known as the modulation propagates through space.

Consider a wave

$$E_x(z,t) = E_m^+ \cos(\omega t - \beta z)$$

Suppose the wave has two frequencies but have equal amplitude, given by

$\omega_1 = \omega_0 + \Delta\omega$

$\omega_2 = \omega_0 - \Delta\omega$

β corresponding to ω_1 & ω_2 are

$\beta_1 = \beta_0 + \Delta\beta$

$\beta_2 = \beta_0 - \Delta\beta$

$\therefore$ Group velocity

$$V_g = \frac{\Delta\omega}{\Delta\beta}$$

18a		Attenuation constant α It's the attenuation of an EM wave propagating through a medium per unit distance from the source. It's the real part of the propagation constant & is measured in nepers/mtr. Phase shift constant β When the EM wave propagates, phase of $\vec{E}$ & $\vec{H}$ rotates. The angular radians rotated per meter travel of the wave is called the phase shift constant. It's the imaginary part of the propagation constant & is measured in radians/mtr. Propagation constant γ EM wave propagate in a sinusoidal fashion. The measure of change in its

amplitude & phase per unit distance is called the propagation constant.

$$\gamma = \alpha + j\beta$$

Intrinsic impedence η

The ratio of amplitudes of $\vec{E}$ & $\vec{H}$ of the waves in either direction is called intrinsic impedence of the material in which wave is travelling. It's denoted by η

$$\eta = \frac{E_m^+}{H_m^+} = -\frac{E_m^-}{H_m^-}$$

$$\eta = \sqrt{\frac{\mu}{\varepsilon}} \ \Omega$$

For free space $\eta_0 = 377 \ \Omega$

Wave length $\lambda = \frac{2\pi}{\beta}$ m

Wave velocity

$$V = f\lambda \quad m/s$$

$$V = \frac{\omega}{\beta} \quad m/s$$

$$V = \frac{1}{\sqrt{\mu\varepsilon}} \quad m/s$$

★ A uniform plane wave propagating in a loss less medium in z direction, the electric field is given by $E(x) = E_0 e^{-j\beta z} \hat{a}_x$. Find Poynting vector.

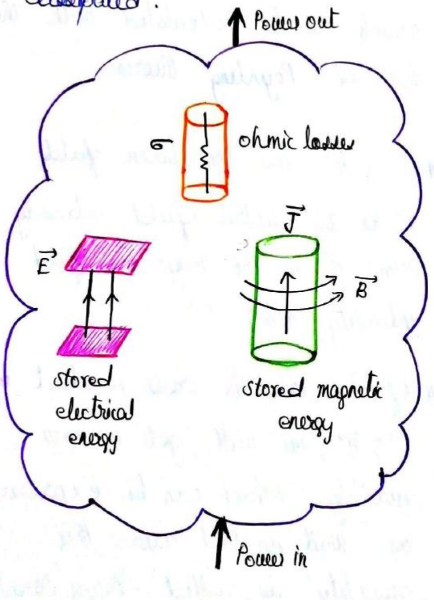
- We have

$$E_x = E_0 e^{-j\beta z} \hat{a}_x$$

18b		Poynting Vector and Poynting Theorem Using EM waves, energy can be transported from transmitter to receiver. The energy stored in electromagnetic fields is transmitted at a certain rate of energy flow which can be calculated with the help of Poynting theorem. # $\vec{E}$ & $\vec{H}$ are the basic fields. $\vec{E}$ is the electric field intensity (V/m), $\vec{H}$ is the magnetic field intensity (A/m). # If we take the cross product of $\vec{E}$ & $\vec{H}$, we will get a new quantity which can be expressed as Watt per Unit area. This quantity is called Power Density Power density, $\vec{P} = \vec{E} \times \vec{H}$ ↓ Poynting vector

The Poynting Theorem states that is based on Law of Conservation of energy in Electromagnetism.

Poynting theorem states that
"The net power flowing out of a given volume V is equal to the time rate of decrease in the energy stored within the volume V minus the ohmic power dissipated".



Suppose $\vec{E} = E_x \hat{a}_x$, $\vec{H} = H_y \hat{a}_y$

$$\vec{P} = \vec{E} \times \vec{H}$$
$$= E_x \hat{a}_x \times H_y \hat{a}_y$$
$$= P_z \hat{a}_z$$

$\vec{E}$, $\vec{H}$ & $\vec{P}$ are mutually perpendicular

$\vec{P}$ is called Poynting vector. $\vec{P}$ is the instantaneous power density vector associated with EM field at a given point.

To get a net power flowing out any surface, $\vec{P}$ is integrated over the same closed surface.

Integral Form of Poynting Theorem

$$\underbrace{\oint_S (\vec{E}\times\vec{H})\cdot \vec{ds}}_{\text{Total power leaving the volume}} = \underbrace{-\frac{\partial}{\partial t}\int_V \left[\frac{1}{2}\varepsilon E^2+\frac{1}{2}\mu H^2\right]dV}_{\text{Rate of decrease in energy stored in EM field}} \underbrace{-\int_V \sigma E^2\, dV}_{\text{ohmic power dissipated}}$$

Proof

Let's consider Maxwell's eqn based on Ampere's Circuital law.

$$\nabla \times \vec{H} = \vec{J_c} + \frac{\partial \vec{D}}{\partial t}$$

$$\vec{J_c} = (\nabla \times \vec{H}) - \frac{\partial \vec{D}}{\partial t} \text{ ---- (A)}$$

$$\nabla \times \vec{H} = \sigma \vec{E} + \varepsilon \frac{\partial \vec{E}}{\partial t}$$

Dot the eqn with $\vec{E}$

$$\vec{E} \cdot (\nabla \times \vec{H}) = \vec{E} \cdot (\sigma \vec{E}) + \vec{E} \cdot \left(\varepsilon \frac{\partial \vec{E}}{\partial t}\right)$$

By vector identity,

$$\vec{E} \cdot (\nabla \times \vec{H}) = \vec{H} \cdot (\nabla \times \vec{E}) - \nabla \cdot (\vec{E} \times \vec{H})$$

$$= \vec{H} \cdot \left(-\mu \frac{\partial \vec{H}}{\partial t}\right) - \nabla \cdot (\vec{E} \times \vec{H})$$

Then

$$\vec{H} \cdot \left(-\mu \frac{\partial \vec{H}}{\partial t}\right) - \nabla \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot (\sigma \vec{E}) + \vec{E} \cdot \left(\varepsilon \frac{\partial \vec{E}}{\partial t}\right)$$

$$\nabla \cdot (\vec{E} \times \vec{H}) = -\vec{E} \cdot \left(\varepsilon \frac{\partial \vec{E}}{\partial t}\right) - \vec{H} \cdot \left(\mu \frac{\partial \vec{H}}{\partial t}\right) - \vec{E} \cdot (\sigma \vec{E})$$

$$\nabla \cdot (\vec{E} \times \vec{H}) = -\frac{\varepsilon}{2} \frac{\partial E^2}{\partial t} - \frac{\mu}{2} \frac{\partial H^2}{\partial t} - \sigma E^2$$

Taking volume integral,

$$\int_V \nabla \cdot (\vec{E} \times \vec{H}) \, dv = -\frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \varepsilon E^2 + \frac{1}{2} \mu H^2\right) dv - \int_V \sigma E^2 \, dv$$

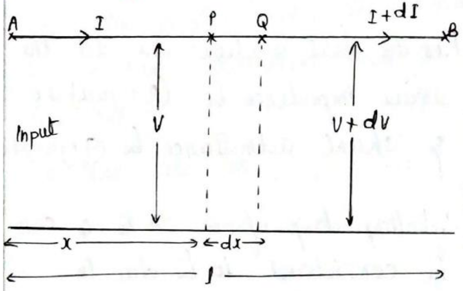
Applying divergence theorem on L.H.S

$$\oint_S (\vec{E} \times \vec{H}) \cdot \vec{ds} = -\frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \varepsilon E^2 + \frac{1}{2} \mu H^2\right) dv - \int_V \sigma E^2 \, dv$$

Hence the proof

19a		

Transmission Line Equation



Consider a short section of TL, PQ of length dx & at a distance 'x' from the sending end A.

At 'P' voltage is 'V' & current is 'I'. At 'Q' voltage is 'V+dV' & current is 'I+dI'.

Let R and L be the loop resistance & inductance per unit length and G & C be the shunt conductance and capacitance per unit length of transmission line.

Assumption

If the length dx is small then,

1. Voltage b/w conductors can be assumed as constant while calculating current.

2. Current in the line is considered constant while calculating voltages.

For the short section dx, let the series impedence be $(R+j\omega L)dx$ & shunt admittance be $(G+j\omega C)dx$.

$\therefore$ Voltage drop from P to Q can be considered to be due to current I flowing through $(R+j\omega L)dx$, the decrease in current from P to Q can be considered due to voltage V applied to the shunt admittance $(G+j\omega C)dx$.

The voltage drop across PQ is given by

$$V-(V+dV) = I(R+j\omega L)dx$$

$$-dV = I(R+j\omega L)dx$$

$$\frac{dV}{dx} = -I(R+j\omega L) \quad \text{--- (1)}$$

Similarly the current difference b/w P & Q is given by,

$$I-(I+dI) = V(G+j\omega C)dx$$

$$-dI = V(G+j\omega C)dx$$

$$\frac{dI}{dx} = -V(G+j\omega C) \quad \text{--- (2)}$$

In order to write each eqn having only one variable, differentiate (1) w.r.t x

$$\frac{d^2V}{dx^2} = -\frac{dI}{dx}(R+j\omega L)$$

Substitute for $\frac{dI}{dx}$

$$\frac{d^2V}{dx^2} = V(G+j\omega C)(R+j\omega L) \quad \text{--- (3)}$$

Differentiate (2) w.r.t x

$$\frac{d^2I}{dx^2} = -\frac{dV}{dx}(G+j\omega C)$$

Substitute for $\frac{dV}{dx}$

$$\frac{d^2I}{dx^2} = I(R+j\omega L)(G+j\omega C) \quad \text{--- (4)}$$

Let $(R+j\omega L)(G+j\omega C) = \gamma^2$

Then (3) & (4) become

$$\frac{d^2V}{dx^2} = \gamma^2 V$$

$$\frac{d^2I}{dx^2} = \gamma^2 I$$

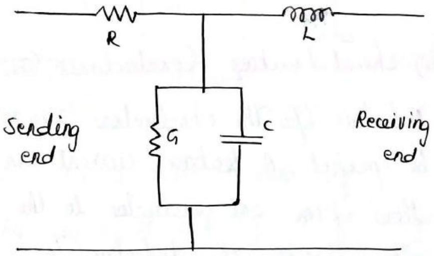
Above eqns are the std differential eqn of transmiss

		is transmission line eqn. General solution of transmission line eqn in exponential form $\frac{d^2V}{dx^2} = \gamma^2 V$ $V = Ae^{-\gamma x} + Be^{\gamma x}$ $\frac{d^2I}{dx^2} = \gamma^2 I$ $I = Ce^{-\gamma x} + De^{\gamma x}$ ABCD are constants $\therefore V = V^+e^{-\gamma x} + V^-e^{\gamma x}$ and $I = I^+e^{-\gamma x} + I^-e^{\gamma x}$ are the solution First term in the eqn of voltage & current shows that there exist voltage & current components travelling towards load impedance. They are called incident wave. The second term gives a wave similar to incident wave but travelling from load end to the source end & is called reflected wave.

19b

Transmission Line Parameters

Primary Characteristics of a Line/ Primary Constants/ Line Parameters



The fig shows the the equivalent circuit of transmission line.

i) Series Resistance (R):

This is a distributed resistor & it is the resistance of both wires together. It shows ohmic loss in the line. Unit is Ω/km.

ii) Series Inductance (L):

Current carrying transmission line has an associated magnetic field which varies with time. Hence it posses the inductance & it's the inductance of both wires together.
Unit is H/km.

iii) Shunt Capacitance (C):

Two conductors separated by a distance situated in a dielectric medium give rise to capacitance which acts in parallel with the line.

Unit is F/km.

iv) Shunt Leakage Conductance (G):

Dielectric b/w the conductors may not be perfect. A leakage current can flow from one conductor to the other through the dielectric. To account for this leakage current, leakage conductance is considered.

Unit is ℧/km.

$Z = R + j\omega L \qquad Y = G + j\omega C$

For a uniform transmission line, R, L, G & C are constants along the length of the line.

For calculation purpose these constants are assumed to be lumped though they are uniformly distributed in transmission line.

Secondary Constants

i) Propagation Constant γ:

It governs the way in which the waves are propagated. Propagation constant is a function of frequency. It's normally expressed in per km. Propagation constant is usually a complex quantity & can be expressed as

$$\gamma = \alpha + j\beta.$$

where α: attenuation constant

β: phase constant.

Phase constant multiplied by the length of transmission line is termed as electrical length of the line.

$$\gamma = \alpha + j\beta$$

$$= \sqrt{(R+j\omega L) + (G + j\omega C)}$$

$$= \sqrt{YZ} \quad m^{-1}$$

ii) Attenuation factor α:

It represents attenuation or reduction of voltage or current along the transmission line.

Unit is Neper/km

iii) Phase shift constant β:

The variation in phase position of voltage or current along the line is determined by it.

Unit is rad/km.

iv) Characteristic Impedence Z_0:

It's the ratio of voltage to current at the input of an infinite line.

$$Z_0 = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$= R_0 + j X_0$$

R_0 & X_0 are the real & imaginary parts of characteristic impedence respectively.

20a	Solution for Loss less lines A transmission line is said to be loss less if conductors of line are perfect and the dielectric medium separating them is loss less. For loss less line, $R = G = 0$ $\alpha = 0$ $\gamma = \alpha + j\beta$ $\therefore \gamma = j\beta$ but $\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$ $j\beta = \sqrt{(j\omega)^2 LC}$ $\cancel{j}\beta = \cancel{j}\omega\sqrt{LC}$ $\beta = \omega\sqrt{LC}$ ~~Sary~~ characteristic impedence $Z_0 = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = R_0 + jX_0$ $\therefore Z_0 = \sqrt{\frac{j\omega L}{j\omega C}} = \sqrt{\frac{L}{C}} = R_0$ The general solution of transmission line eqn is $V = V^+ e^{-\gamma x} + V^- e^{\gamma x}$

	$I = I^+ e^{-\gamma x} + I^- e^{\gamma x}$ Substituting for $\gamma = j\beta$ $V = V^+ e^{-j\beta x} + V^- e^{j\beta x}$ $I = I^+ e^{-j\beta x} + I^- e^{j\beta x}$
20b	$\Gamma_R = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{150-50}{150+50} = 0.5$ $VSWR = \frac{1+\Gamma_R}{1-\Gamma_R} = \frac{1+0.5}{1-0.5} = 3$ $VSWR = \frac{V_{max}}{V_{min}} = 3$, $V_{min} = \frac{V_{max}}{3} = \frac{30}{3} = 10V$